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MILITARY SPECIFICATION

AMPLIFIER, AUDIO FREQUENCY AM–1098/MSE–2 (FOR MISSILES)

1. SCOPE

1.1 This specification covers one type of amplifier designated as Amplifier, Audio Frequency AM–1098/MSE–2.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

UU–T–106 —Tape, Pressure-Sensitive Adhesive, Masking, Paper.
UU–T–111 —Tape, Paper, Gummed, Sealing and Securing.
PPP–B–585 —Boxes, Wood, Wirebound.
PPP–B–601 —Boxes, Wood, Cleated-Plywood.
PPP–B–621 —Boxes, Wood, Nailed and Lock-Corner.
PPP–B–636 —Box, Fiberboard.
PPP–B–640 —Boxes, Corrugated, Triple Wall, 350 Pound Maximum Weight.
PPP–T–76 —Tape, Pressure-Sensitive Adhesive, Paper, Water Resistant.

MILITARY

MIL–P–116 —Preservation, Methods of.
MIL–B–131 —Barrier Material, Water Vaporproof, Flexible.
MIL–T–152 —Treatment, Moisture- and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment.
MIL–T–945 —Test-Equipment, for use with Electronic Equipment: General Specification.
MIL–E–5272 —Environmental Testing, Aeronautical and Associated Equipment, General Specification for.
MIL–C–7769 —Cushioning Material, Bound.

STANDARDS

MILITARY

MIL–STD–129 —Marking for Shipment and Storage.
MIL–STD–130 —Identification Marking of U. S. Military Property.
MIL–STD–202 —Test Methods for Electronic and Electrical Component Parts.
MIL–STD–252 —Wired Equipment, Classification of Visual and Mechanical Defects for.

FSC 1430

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DRAWINGS

ORDNANCE CORPS

9155139 —Amplifier, Audio Frequency, AM–1098/MSE–2.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 **Other publications.** The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Uniform Freight Classification Rules and Container Specifications for Rail Shipment.

(Application for copies should be addressed to the Uniform Freight Classification Committee, 202 Union Station, Chicago 6, Illinois).

National Motor Freight Classification Rules and Container Specifications for Truck Shipments.

(Application for copies should be addressed to the National Classification Board, 1424 Sixteenth Street, N. W., Washington 6, D. C.).

3. REQUIREMENTS

3.1 **Preproduction sample.** When a preproduction sample is required (see 6.2), it shall meet all the requirements of this specification. The preproduction sample shall be manufactured using the same methods proposed for the manufacture of subsequent production lots.

3.2 **Material.** All materials used in the construction of each amplifier shall be in accordance with Drawing 9155139 and applicable drawings and specifications referenced thereon.

3.3 **Design and construction.** Design and construction of the amplifier shall be in accordance with Drawing 9155139 and applicable drawings and specifications referenced thereon.

3.4 **Electrical.**

3.4.1 *Ground potential.* The design and construction of the amplifier shall be such that all external metallic parts are at ground potential at all times.

3.4.2 *Dielectric strength.* There shall be no puncture of insulation, no sparkover, or any other evidence of breakdown in the dielectric when the insulated areas of the amplifier are subjected to an alternating current (ac) root mean square (rms) potential of 500 volts plus twice the rated voltage at a frequency of 60 cycles per second (cps) for a period of not less than 3 seconds nor more than 5 seconds.

3.4.3 *Insulation resistance.* Unless otherwise specified on applicable drawings, all insulation shall have a minimum resistance of 100 megohms between isolated circuits and between ungrounded circuits and ground.

3.5 **Performance and product characteristics.**

3.5.1 *Input voltages.* The amplifier shall be capable of operating satisfactorily when supplied with the following input voltages:

(a)	Positive 250 volts direct-current (dc).	Terminal P1–8 and ground
(b)	Negative 250 volts dc.	Terminal P1–2 and ground
(c)	Negative 28 volts dc.	Terminal P1–4 and ground
(d)	6.3 volts ac, 400 cps, single phase.	Terminal P1–14 and P1–15

3.5.1.1 *Power consumption.* Power consumption shall be such that the current shall not exceed that specified in Table I.

TABLE I. *Power consumption*

Voltage	Current
Positive 250 volts dc	20 milliamperes (ma)
Negative 250 volts dc	10 ma
Negative 28 volts dc	500 ma
6.3 volts, ac, 400 cps	1.2 amperes

3.5.2 *Zero adjustment.* With a zero voltage input, the amplifier shall be capable of being adjusted by R5 for a balanced output.

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3.5.2.1 *Stability.* With a balanced output, the voltage variation shall be not greater than plus or minus 1 volt.

3.5.3 *Positive 1 volt input.* With a positive 1 volt input to the amplifier, the output voltage shall be negative 25 volts plus or minus 1.5 volts.

3.5.4 *Negative 1 volt input.* With a negative 1 volt input to the amplifier, the output voltage shall be positive 25 volts plus or minus 1.5 volts.

3.5.5 *Positive ¼-volt input.* With a positive ¼ volt input to the amplifier, the output shall be negative 8 volts plus or minus 2 volts.

3.5.6 *Negative ¼-volt input.* With a negative ¼ volt input to the amplifier, the output shall be positive 8 volts plus or minus 2 volts.

3.5.7 *Output waveform.* With a 16 cps oscillator input to the amplifier, the oscillator shall be adjusted to the minimum voltage which will give an amplifier output waveform similar to Figure 1. The oscillator input voltage shall be not greater than 1 volt.



FIGURE 1. *Waveform.*

3.6 **Environmental.**

3.6.1 *Operating temperature.* The amplifier shall operate satisfactorily within ambient temperatures ranging from minus 40° centigrade (C) to plus 51.7° C. (minus 40° to 125° Fahrenheit (F)).

3.6.2 *Nonoperating temperature.* The amplifier shall not be damaged nor shall its performance be impaired by storage on transportation at temperatures ranging from minus 53.8° C. to plus 60° C. (minus 65° to plus 140° F.).

3.6.3 *Vibration.* The amplifier shall not be damaged nor shall its performance be impaired when subjected to vibrating frequencies ranging from 10 to 55 cps having a total excursion of 0.0156 inch plus or minus 10 percent.

3.6.4 *Shock.* The amplifier shall not be damaged nor shall its performance be impaired by shocks that normally occur during operating and nonoperating conditions.

3.6.5 *Relative humidity.* The amplifier shall not be damaged nor shall its performance be impaired when subjected to a relative humidity of 90 percent.

3.6.6 *Fungus.* The amplifier shall be treated in accordance with Specification MIL–T–152. The unit shall not be damaged nor shall its performance be impaired when subjected to the fungus treatment.

3.6.7 *Salt-laden air.* The amplifier shall not be damaged nor shall its performance be impaired by salt-laden air as encountered in coastal service or in ocean transportation.

3.6.8 *Altitude.* The amplifier shall operate satisfactorily at any altitude from sea level to 10,000 feet above sea level and shall not be damaged from transportation by air up to and including 30,000 feet (nonoperating).

3.7 **Details of components.** Components of the amplifier shall conform to Drawing 915-5139 and applicable drawings and specifications referenced thereon.

3.8 **Dimensions.** Dimensions and tolerances of the amplifier shall be in accordance with Drawing 9155139 and applicable drawings referenced thereon.

3.9 **Color and finish.** Color and finish of the amplifier shall be in accordance with Drawing 9155139 and applicable drawings and specifications referenced thereon.

3.10 **Product marking.** Product marking of the amplifier shall be in accordance with Drawing 9155139 and applicable drawings and specifications referenced thereon and Standard MIL–STD–130.

3.11 **Workmanship.** All components shall be fitted properly and firmly into their respective positions. All mechanical joints and electrical connections shall be mechanically secure and electrically sound. All controls

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shall operate freely without slipping, sticking, or binding. Electrical connections shall be flexible between shock mounted components and their foundations. Surfaces of glazed ceramic components shall be smooth, uniform and free from cracks. There shall be no evidence of corrosion, burrs, roughness, sharp edges, or other defects which could adversely affect its use or suitability for the purpose intended.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction sample. The preproduction sample shall be subjected to all examinations and tests specified herein. Unless otherwise specified, the Government will perform the examinations and tests at the contractor's plant. If the preproduction sample does not meet all requirements of this specification, it will be rejected and returned to the contractor. Subsequent units or lots as applicable will not be considered for acceptance until approval of the preproduction sample has been obtained.

4.3 Inspection. Unless otherwise specified by the contract or contracting officer (see 6.2), inspection for examination and testing shall be 100 percent of production (see 4.8).

4.4 Lot size. Lot size shall not exceed that quantity of amplifiers produced in one continuous operation by the same manufacturer and manufactured in accordance with the same drawing, drawing revision, specification, and specification revision.

4.5 Examination. The amplifier shall be given a visual and tactile examination for defects to determine compliance with the requirements of 3.2, 3.3, 3.7, 3.8, 3.9, 3.10, and 3.11. Standard MIL–STD–252 shall be used where applicable in this examination.

4.6 Test equipment. Test equipment shall meet the minimum requirements of the equipment listed below and shall be used as indicated in performing the tests. In the test setup and test procedures, the equipment is referenced to by item number. The equipment may be mounted separately or in any combination (such as consoles, panels, and multipurpose instruments).

Item	*Description*
1	Power supply, 6.3 volts ac plus or minus 0.5 volt, 400 cps, 1.2 amperes, single phase.
2	Power supply, negative 28 volts dc plus or minus 2 percent, 500 ma.
3	Power supply, positive 250 volts dc plus or minus 2 percent, 20 ma.
4	Power supply, negative 250 volts, dc plus or minus 2 percent, 10 ma matched within 1 percent of item 3.
5	Ampere meter, dc, one to zero to one milliampere scale.
6	Resistor, 2895 ohms plus or minus 1 percent.
7	Resistor, 27,000 ohms plus or minus 1 percent.
8	Resistor, 250 ohms plus or minus 1 percent (2 required).
9	Resistor, 750 ohms plus or minus 1 percent (2 required).
10	Resistor, 249,000 ohms plus or minus 1 percent.
11	Resistor, 100 ohms plus or minus 1 percent, 20 watts.
12	Switch, rotary, 5 position.
13	Switch, single-pole, single-throw.
14	Oscillator, 16 cps.
15	Oscilloscope, 5-inch screen, internal or external triggered sweep, internal sweep variable from 2 to 30,000 cps, vertical deflection 0.01 volt per inch, horizontal deflection 0.1 volt per inch, frequency response 0 to 100,000 cps, input impedance 2 megohms.
16	Voltmeter, portable, ac rectifier type, 10 to 150,000 cps, 0.001 volt to 100 volts plus or minus 2 percent.
17	Multimeter, ac range 0 to 6.3 volts; dc ranges 0 to negative 28 volts, and 0 to 250 volts (positive and negative); dc ma scale from 0 to 500 ma; ohms scale of 0 to 10 megohms.
18	Ampere meter, ac range of 0 to 1.2 amperes plus or minus 2 percent.

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In inspecting the suppliers inspection equipment, the Government inspector will determine that the supplier has available, and utilizes correctly, gaging, measuring, and test equipment of required accuracy and precision and that the instruments are of proper type and range to make measurements within the required accuracy. The supplier will have available a set of master gages, standards, and appropriate instruments for regularly scheduled calibration of his inspection equipment. Records of such regularly scheduled calibration will be maintained by the supplier and made available for review by the Government. The calibration of gages, standards, and instruments will be periodically checked by authorized Government personnel.

4.7 **Test setup.** Unless otherwise specified, test equipment and items shall be connected to the amplifier as indicated in figure 2 for the performance and product characteristics tests. Prior to conducting the tests, the amplifier shall be given a 5-minute warm-up period.

4.8 **Test methods and procedures.** The following tests shall be performed on 100 percent of production with the exception of the power consumption and environmental tests which shall be performed on the preproduction sample only. The tests specified in 4.8.1 shall be performed prior to the application of power. Tests specified in 4.8.2 shall be performed in the sequence indicated.

4.8.1 *Electrical.*

4.8.1.1 *Ground potential.* All external metallic parts shall be measured to terminal P1–13 with item 17 to determine compliance with 3.4.1.

4.8.1.2 *Dielectric strength.* The amplifier shall be tested in accordance with Standard MIL–STD–202, method 301 and shall conform to the requirements specified in 3.4.2.

4.8.1.3 *Insulation resistance.* Immediately after satisfactorily withstanding the tests for dielectric strength, the amplifiers shall be given the insulation resistance test in accordance with Standard MIL–STD–202, method 302, condition "A" and shall conform to the requirements specified in 3.4.3.

4.8.2 *Performance and product characteristics.*

4.8.2.1 *Input voltages.* Using item 17, measure the output voltages of items 1, 2, 3, and 4 to insure compliance with 3.5.1.

4.8.2.1.1 *Power consumption.* Using items 17 and 18, measure the current of items 1, 2, 3, and 4 to insure compliance with 3.5.1.1.

4.8.2.2 *Zero adjustment.* Item 12 shall be set to the "Zero" position and measured by item 17. Item 5 shall be connected as indicated in figure 2 and the reading shall be on scale. Resistor R5, in the amplifier, shall be adjusted for a zero (center scale) reading as indicated on item 5. Item 13 shall be operated, so as to short out item 7, and R5 shall be readusted to a zero reading to insure compliance with 3.5.2.

4.8.2.3 *Stability.* Upon conclusion of the test in 4.8.2.2 the voltage variation shall be not greater than plus or minus 1 volt as measured by item 16 to insure compliance with 3.5.2.1.

4.8.2.4 *Positive 1 volt input.* With item 12 set to "Plus 1" position, and measured with item 17, the output voltage shall be negative 25 volts plus or minus 1.5 volts as indicated on item 17 to insure compliance with 3.5.3.

4.8.2.5 *Negative 1 volt input.* With item 12 set to "Minus 1" position and measured with item 16, the output voltage shall be positive 25 volts plus or minus 1.5 volts as indicated on item 17 to insure compliance with 3.5.4.

4.8.2.6 *Positive ¼ volt input.* With item 12 set to "Plus ¼" position and measured with item 17, the output shall be negative 8 volts plus or minus 2 volts as indicated on item 17 to insure compliance with 3.5.5.

4.8.2.7 *Negative ¼ volt input.* With item 12 set to "Minus ¼" position and measured with item 17, the output shall be positive 8 volts plus or minus 2 volts as indicated on item 17 to insure compliance with 3.5.6.

4.8.2.8 *Output waveform.* Connect items 1,

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2, 3, and 4 as in figure 2 and items 14, 15, and 16 as in figure 3. Item 14 shall be adjusted until the waveform displayed on item 15 is similar to figure 1. The voltage indicated on item 16 shall be not greater than 1 volt to determine compliance with 3.5.7.

4.8.3 *Environmental.*

4.8.3.1 *Temperature (operating and non-operating).* The amplifier shall be tested in accordance with the high-low temperature test of Specification MIL–E–5272, procedure II, except that the extremes of temperature shall be in accordance with 3.6.1 and 3.6.2.

4.8.3.2 *Vibration.* The amplifier shall be tested in accordance with the vibration test of Specification MIL–T–945, test 2, except that the amplifier shall be in a non-operating condition and the total excursion shall be 0.0156 inch plus or minus 10 percent, to determine compliance with 3.6.3.

4.8.3.3 *Shock.* Tilt the amplifier, nonoperating, up through an angle of 30 degrees or six inches, whichever is less, using one edge of the assembly as an axis, and permit the amplifier to drop back freely to the horizontal position on a 2-inch-thick hardwood surface. Repeat, using other practicable edges of the same horizontal face as axes, for a total of four drops. There shall be no failures due to the applied shocks (see 3.6.4).

4.8.3.4 *Relative humidity.* The amplifier shall be tested in accordance with the humidity test of Specification MIL–E–5272, procedure III, except that the relative humidity shall be 90 percent. The amplifier shall conform to the requirements of 3.6.5 upon conclusion of the humidity test.

4.8.3.5 *Fungus.* The amplifier shall be tested in accordance with the fungus test of Specification MIL–E–5272, procedure I, except that no evidence of fungus growth shall be present upon conclusion of the tests (see 3.6.6).

4.8.3.6 *Salt-laden air.* The amplifier shall be tested in accordance with the salt-spray test of Specification MIL–E–5272, procedure I. The amplifier shall conform to the requirements of 3.6.7 upon conclusion of the salt-spray test.

4.8.3.7 *Altitude.* The amplifier shall be tested in accordance with the temperature—altitude test of Specification MIL–E–5272, procedure II. The altitude shall be as specified in 3.6.8 and the unit shall be non-operating for the 30,000 foot test. The temperature shall be within 5 degrees C of the extremes specified in 3.6.1 and 3.6.2. Upon conclusion of the altitude tests, the amplifier shall operate satisfactorily.

4.9 Preservation, packaging and packing.

4.9.1 *Packaging.* Tests for interior packages shall be in accordance with the applicable requirements of Specification MIL–P–116.

4.9.2 *Packing.* When specified (see 6.2) a level A pilot pack shipping container shall be subjected to the free fall drop test as specified in Specification MIL–P–116.

5. PREPARATION FOR DELIVERY

5.1 *Preservation and packaging.* Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 *Level A.*

5.1.1.1 *Cleaning.* The amplifier shall be cleaned in accordance with Specification MIL–P–116, process C–1.

5.1.1.2 *Drying.* The amplifier shall be dried in accordance with any applicable procedure of Specification MIL–P–116.

5.1.1.3 *Unit packaging.* Each amplifier shall be unit packaged, method IIb of Specification MIL–P–116 as prescribed herein. Each amplifier shall be immobilized in a snug fitting container conforming to Specification PPP–B–636, style RSC, class 1. Desiccant shall be computed in accordance with formula I of Specification MIL–P–116 and shall be distributed, insulated from contact with the amplifier and secured within the container. Container corners shall be blunted prior to insertion in a barrier bag made of material conforming to Specification MIL–B–131, class

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1. The bagged unit shall then be inserted in a snug fitting container conforming to Specification PPP–B–636, style RSC, class 2, or MIL–B–43014, from 4, lined with one inch thick cushioning material conforming to Specification MIL–C–7769, type I or II.

5.1.1.4 *Container closure.* Containers within water vaporproof barrier material may be closed with adhesive or two strips of tape applied the length of the top and bottom seams. Tape shall conform with Specification UU–T–111 or UU–T–106. The outer container shall be closed with two strips of 2 inch width tape conforming to Specification PPP–T–76 applied the length of the seams. When the bottom seam is closed with staples or adhesives, only one strip of tape the length of the top seam is required. When the unit package becomes the shipping container, closure shall be in accordance with the appendix of Specification PPP–B–636, except that tape conforming to PPP–T–76 shall be used and the manufacturers joint or seam shall be sealed.

5.1.2 *Level C.* Unless otherwise specified, level C preservation and packaging shall be the same as level A except that desiccant and barrier material shall be omitted. Preservation shall be method III of Specification MIL–P–116.

5.2 *Packing.* Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 *Level A.* A quantity of unit packages shall be immobilized in an exterior container conforming to Specification PPP–B–585, class 3; PPP–B–601, overseas type; PPP–B–621, style 4, class 2; PPP–B–636, class 2; or PPP–B–640, class 2. The containers shall be closed and strapped in accordance with the appendix of the applicable container specification. The gross weight shall not exceed 200 pounds. When fiberboard containers are the selected container, dimensional and weight limitations shall be complied with.

5.2.2 *Level B.* Amplifiers packaged as specified in 5.1.1 shall be packed the same as for level A except that selected wood or wood-cleated containers shall be of a non-weather resistant or domestic type, grade, or class.

5.2.3 *Level C.* Amplifiers packaged as prescribed herein shall be packed in containers that conform to the rules and regulations applicable to mode of transportation. Containers shall comply with Uniform Freight Classification Rules and Container Specifications for rail shipment and National Motor Freight Classification Rules and Container Specification for truck shipments.

5.3 *Marking.* In addition to any special marking required by the contract or order (see 6.2), interior packages and exterior shipping containers shall be marked in accordance with Standard MIL–STD–129.

6. NOTES

6.1 *Intended use.* The Amplifier, Audio Frequency AM–1098/MSE–2 furnishes up to plus or minus 28 volts dc driving voltage to a servo motor in the missile, to preset the roll gyro, under control of plus or minus 0.5 volts maximum dc error voltage. (NIKE Missile System).

6.2 *Ordering data.* Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Whether a preproduction sample is required (see 3.1).
(c) Number of preproduction samples required (see 3.1).
(d) Where the preproduction sample tests shall be performed (see 4.2).
(e) Inspection other than specified (see 4.3).
(f) Selection of applicable levels of preservation, packaging, and packing (see 5.1 and 5.2).
(g) Whether a pilot pack is required.
(h) Whether special marking is required (see 5.3).
(i) Applicable stock number.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may

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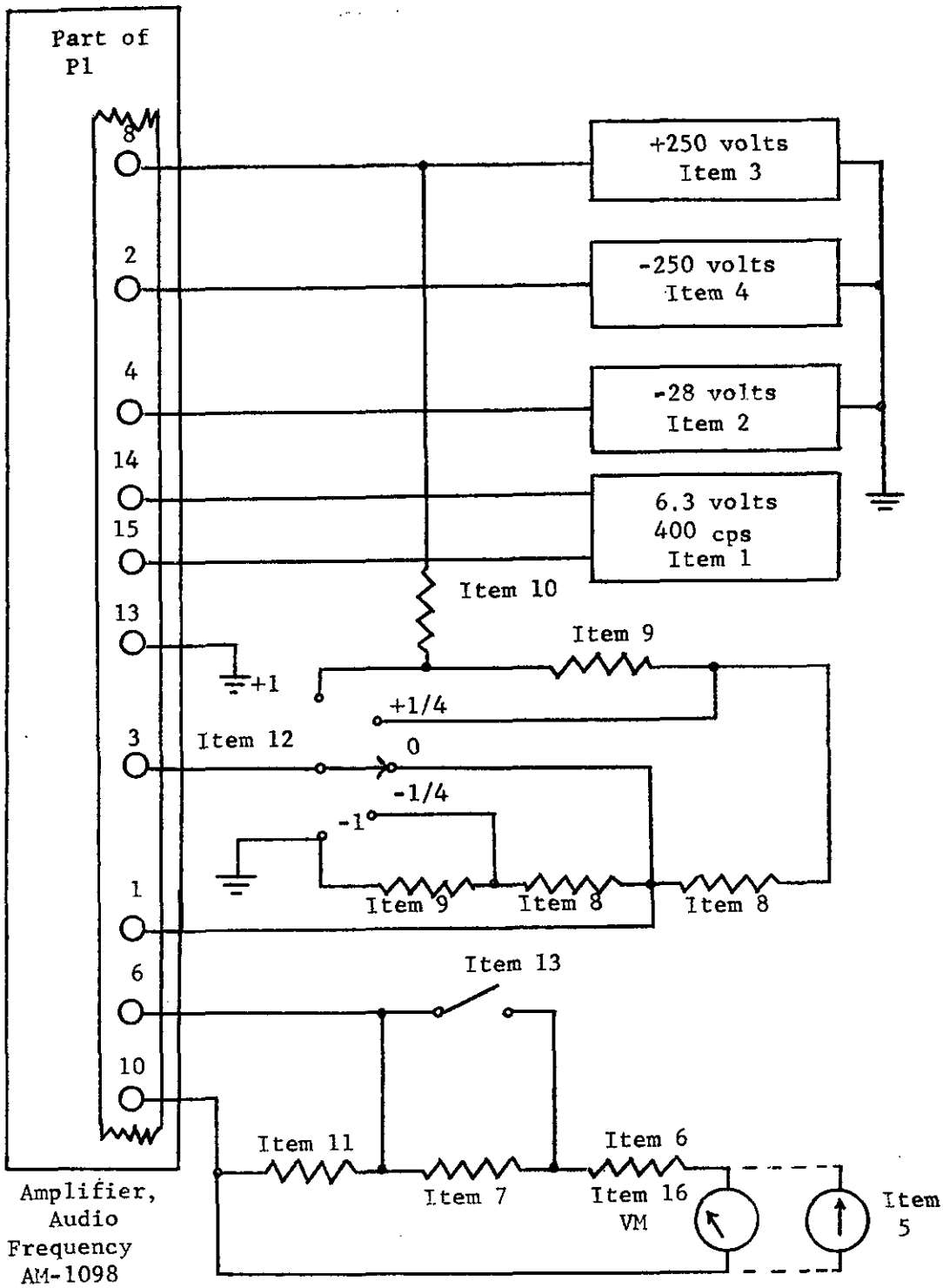


FIGURE 2.

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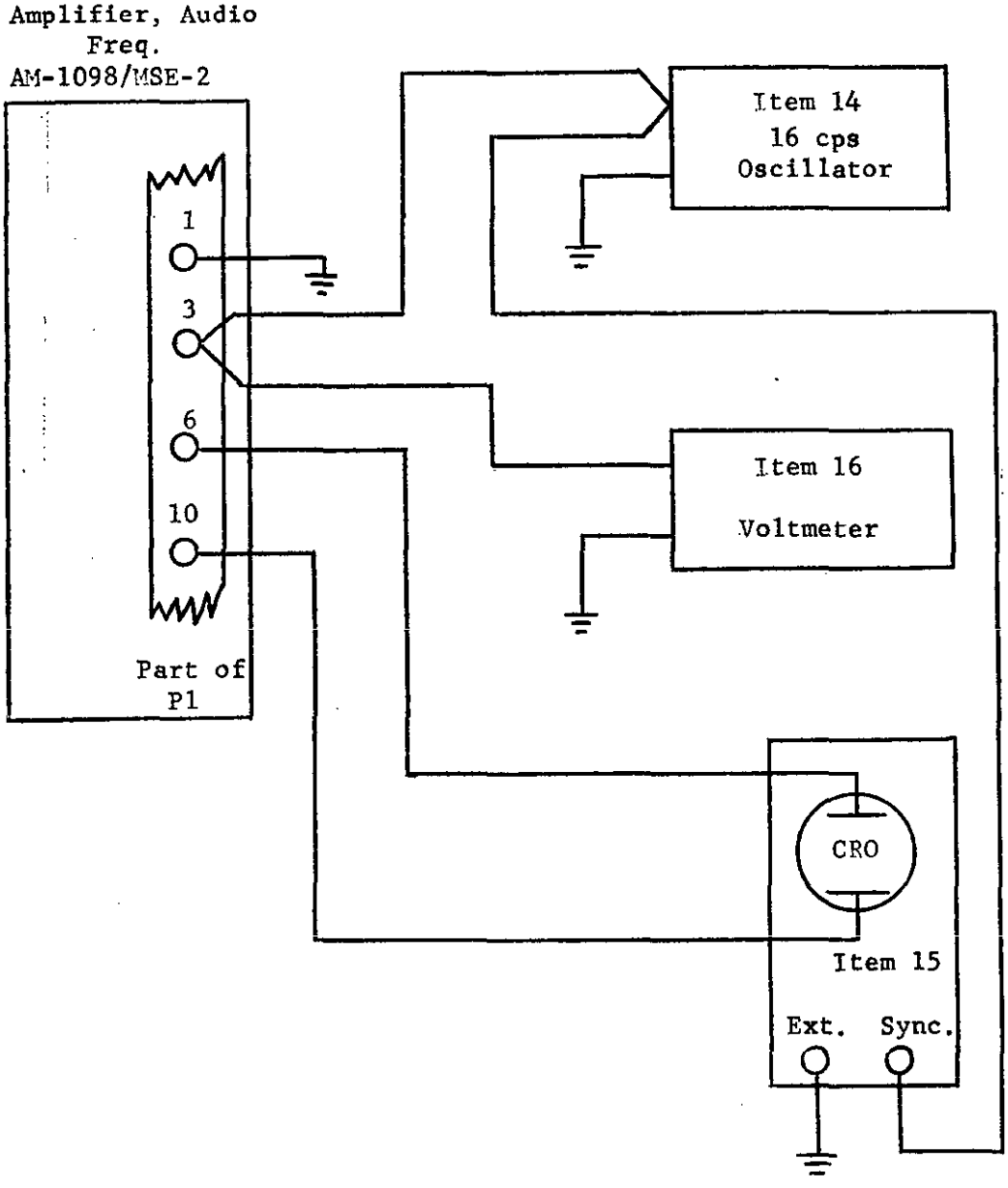


FIGURE 3. *Test connections.*

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